

**DEPARTMENT OF MATHEMATICS & STATISTICS**  
**MTH 211: THEORY OF STATISTICS**  
**Midsemester Exam**  
**Session 2025-26 Semester II**

**Maximum Marks: 60**

**Time: 120 minutes**

1. Let  $f(x, y|\theta_1, \theta_2)$  be the bivariate pdf for the uniform distribution on the rectangle with lower left corner  $(0, 0)$  and upper right corner  $(\theta_1, \theta_2)$  in  $\mathbb{R}^2$ , where  $\theta_1, \theta_2 > 0$ . Let  $(X_1, Y_1), \dots, (X_n, Y_n)$  be a random sample from this pdf.
  - (a) Find the jointly minimal sufficient statistic (MSS) for  $(\theta_1, \theta_2)$ . (4 marks)
  - (b) For the special case  $\theta_1 = \theta_2 = \theta$ , prove or disprove that the statistic obtained in (a) is complete. (4 marks)
2. As a statistical researcher at Professor Oak's lab, you are tasked with analysing the properties of various Pokémon populations. Answer the following:
  - (a) A wild Ditto is observed mimicking objects near a power plant. Its distance  $X$  from the plant follows a distribution with the pdf:

$$f(x|\theta) = \frac{2x}{\theta^2}, \quad 0 \leq x \leq \theta$$

- where  $\theta > 0$  is an unknown parameter. You observe  $n$  independent positions  $X_1, X_2, \dots, X_n$ . Prove or disprove that an MSS, say,  $T$  is complete. (5 marks)
- (b) The weight-gain of a Snorlax follows a  $N(\mu, \sigma^2)$  distribution, where both  $\mu \in \mathbb{R}$  and  $\sigma^2 > 0$  are unknown. Based on a random sample of  $n$  weight gains, construct the UMVUE for the parameter  $g(\mu) = \mu^3$ . (5 marks)
3. Let  $X_1, X_2, \dots, X_n$  be a random sample from a Gamma distribution,  $X \sim \text{Gamma}(\alpha, \beta)$ . Find the MOME for the parameter vector  $(\alpha, \beta)$ . (7 marks)
4. The background electrical noise  $X$  (in millivolts) in a high-precision sensor at an IITK lab is modeled by a normal distribution  $N(0, \sigma^2)$ . The sensor is calibrated such that there is no DC offset (mean is zero). A technician records  $n$  independent noise readings  $X_1, X_2, \dots, X_n$ .
  - (a) Find the CRLB for variance of an unbiased estimator of  $\sigma^2$ . (4 marks)
  - (b) A student proposes using the sample variance  $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$  to estimate  $\sigma^2$ . Prove or disprove that  $S^2$  attains the CRLB. (3 marks)
5. Let  $X_1, X_2, \dots, X_n$  be a sample from the inverse Gaussian pdf:

$$f(x|\mu, \lambda) = \left( \frac{\lambda}{2\pi x^3} \right)^{1/2} \exp \left\{ -\lambda(x - \mu)^2 / (2\mu^2 x) \right\}, \quad x > 0.$$

Obtain the MLE of  $\mu$  and  $\lambda$ . (12 marks)

6. Let  $X_1, X_2, \dots, X_n$  be a sample from a distribution with pdf:

$$f(x|\theta) = \theta \exp(-\theta x), \quad x \geq 0,$$

where  $\theta > 0$  is an unknown parameter. Obtain MLE of  $\theta$ . Show that the MLE is a consistent estimator and obtain its asymptotic distribution. (8 marks)

7. Let  $X_1, X_2, \dots, X_n$  be a random sample from a uniform  $(0, \theta)$  distribution, where  $\theta > 0$  is an unknown parameter. Using the same pivot function, obtain two 2-sided  $100(1 - \alpha)\%$  confidence intervals for  $\theta$ . (8 marks)

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